ModelIt!

Student Activity Pack

Can We Run Out of Rare Earth Metals?

Mining, Recycling, and the Hidden Cost of Modern Technology

Grade 07L2 | NGSS MS-ESS3-1, MS-ESS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think is inside your phone that makes it work? Where do those materials come from?**

2. **What happens to old electronics when you throw them away?**

3. **Draw what you think happens to the minerals inside a phone after it's discarded.**

4. **Do you think Earth has an unlimited supply of the materials we use for technology? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Mining Rate | Recycling Rate | Available Supply | Market Price | Substitution Incentive

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High Mining, Low Recycling

Aggressive extraction, minimal recycling

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Balanced Approach

Moderate mining with high recycling

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Depletion Scenario

Low mining (exhausted deposits), low recycling

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Geopolitical Risk — The vulnerability of supply chains when a single country controls the majority of mining or processing, creating risk of trade disruptions

Technological Demand Growth — The increasing global demand for rare earth metals driven by electric vehicles, renewable energy, smartphones, and defense technology

Environmental Mining Cost — The ecological damage from rare earth extraction including toxic waste, water contamination, habitat destruction, and radioactive byproducts

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Every phone, computer, and electric car uses rare earth metals. If we keep mining them at current rates without recycling, will we run out? Use evidence from your simulation to support your answers.

1. **Why is it misleading to say rare earth metals are 'rare'? How does your model show that the real problem is concentration and extraction, not total amount?**

2. **How does the relationship between supply, price, and substitution create a natural economic feedback loop?**

3. **Why is recycling rare earths from electronics so much more challenging than recycling aluminum cans?**

4. **What would your model predict if a major geopolitical event suddenly cut off 60% of the world's rare earth supply?**

5. **How does the concept of a 'circular economy' differ from the traditional 'mine-use-discard' model, and what does your simulation show about its benefits?**

8 STEM Challenge: Design a Rare Earth Recycling Program

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YOUR ENGINEERING MISSION

Create a community electronics recycling program that maximizes rare earth metal recovery, using your simulation model data to justify the program's design and expected impact.

THE SCENARIO

Your city council wants to reduce electronic waste and recover valuable rare earth metals from discarded devices. Your team has been asked to design a recycling program that collects old electronics, extracts rare earth metals, and returns them to the supply chain, using your model data to project the program's impact.

Engineering Design Thinking

1. WHAT COLLECTION METHODS WILL YOUR PROGRAM USE (DROP-OFF, CURBSIDE, MAIL-IN)?
2. HOW WILL YOU SEPARATE AND EXTRACT RARE EARTH METALS FROM MIXED ELECTRONIC WASTE?
3. WHAT MODEL DATA WILL YOU USE TO PROJECT THE PROGRAM'S IMPACT ON LOCAL SUPPLY AND PRICES?
4. HOW WILL YOU MEASURE THE PROGRAM'S SUCCESS IN TERMS OF MATERIAL RECOVERED AND WASTE DIVERTED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials scientists study the properties and applications of metals, ceramics, polymers, and composites. They develop new materials and recycling processes for industries including technology, aerospace, and renewable energy, earning $70,000-$130,000/year.